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Talk Kindly

Offensive Language Detection

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• BSc Computer Science 2026

1 ABSTRACT

Offensive language is increasingly common in online communication and can negatively affect users and digital communities (Schmidt and Wiegand, 2017). Talk Kindly is a web application developed to detect offensive language using a lexicon based NLP approach. The system processes user text through normalisation, tokenisation, and keyword matching to identify harmful expressions. Testing on 70 sentences produced 90% accuracy and 100% recall after lexicon refinement. The results show that a small, carefully designed lexicon can perform effectively for focused offensive language detection tasks.

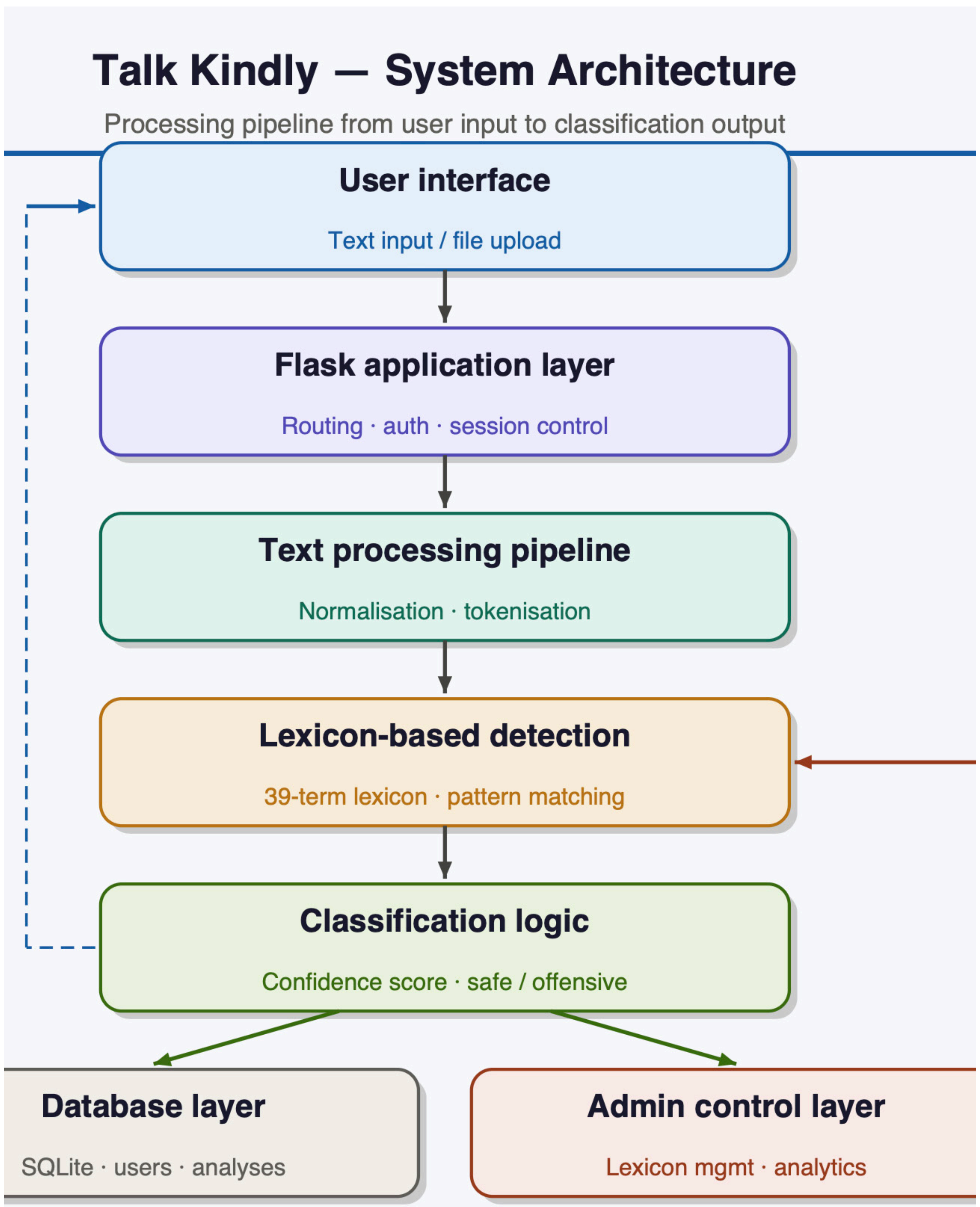
2 AIMS & OBJECTIVES

Aim: to design and implement a web based application that detects offensive language in user submitted text using a transparent lexicon based NLP approach.

- 01 Develop a rule based detection pipeline using normalisation, tokenisation, and lexical matching
- 02 Build and refine a manually constructed offensive language lexicon of 39 terms
- 03 Design a simple web interface for analysing user submitted text
- 04 Evaluate system performance using accuracy, precision, recall, and F1-score
- 05 Compare detection results against the HurtLex lexical resource

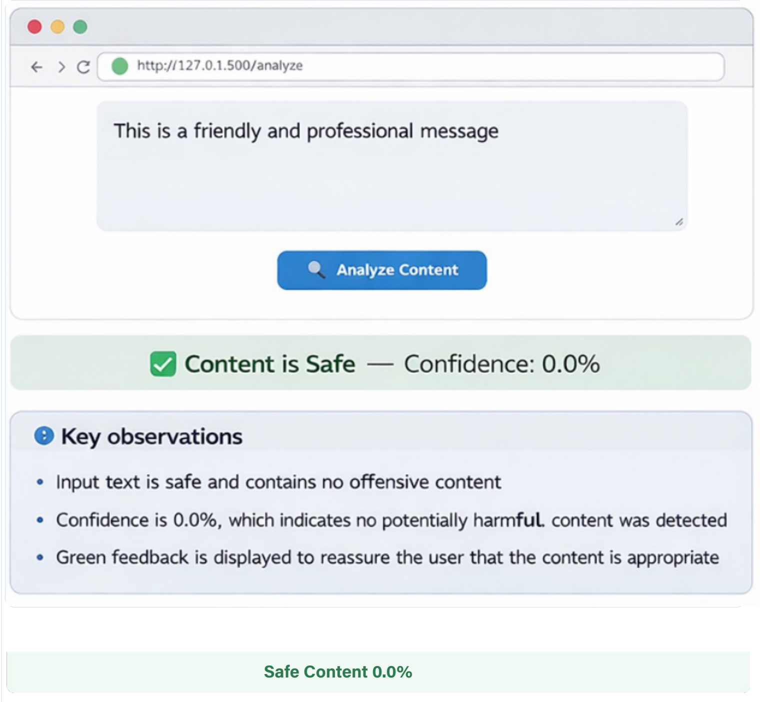
3 SYSTEM ARCHITECTURE

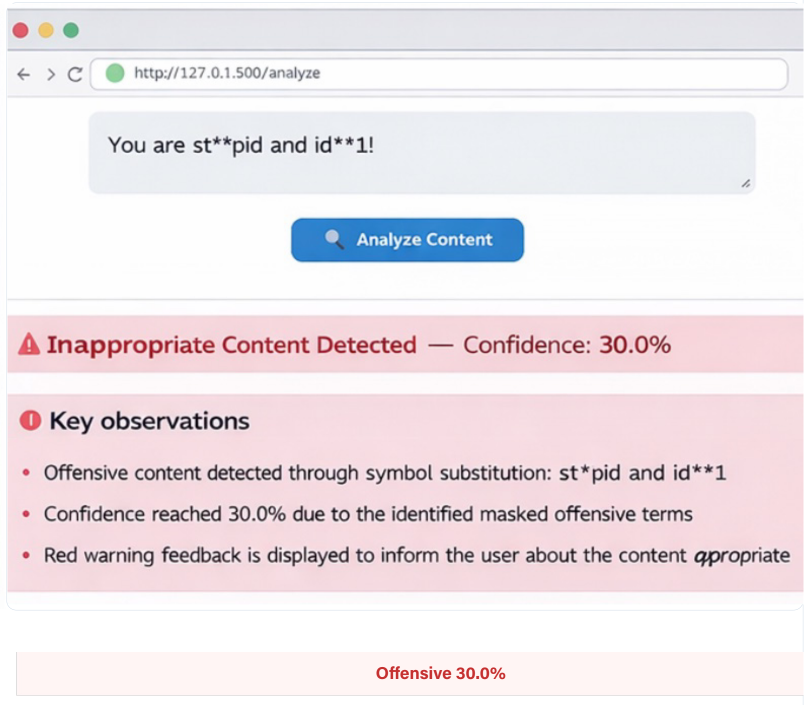
The Talk Kindly system follows a rule based NLP pipeline integrated within a web application interface. User submitted text is first processed through normalisation and tokenisation before being analysed using lexical matching against a manually constructed offensive language lexicon. The classification result is then displayed through the web interface and stored in a SQLite database for history tracking and administrative review.



4 IMPLEMENTATION

The system was developed as a local web based application using Python and Flask. Users can submit text or upload documents for analysis through an interactive interface. The detection pipeline identifies offensive expressions and returns colour coded feedback, showing safe content in green and offensive content in red, together with a confidence score and highlighted matched patterns.



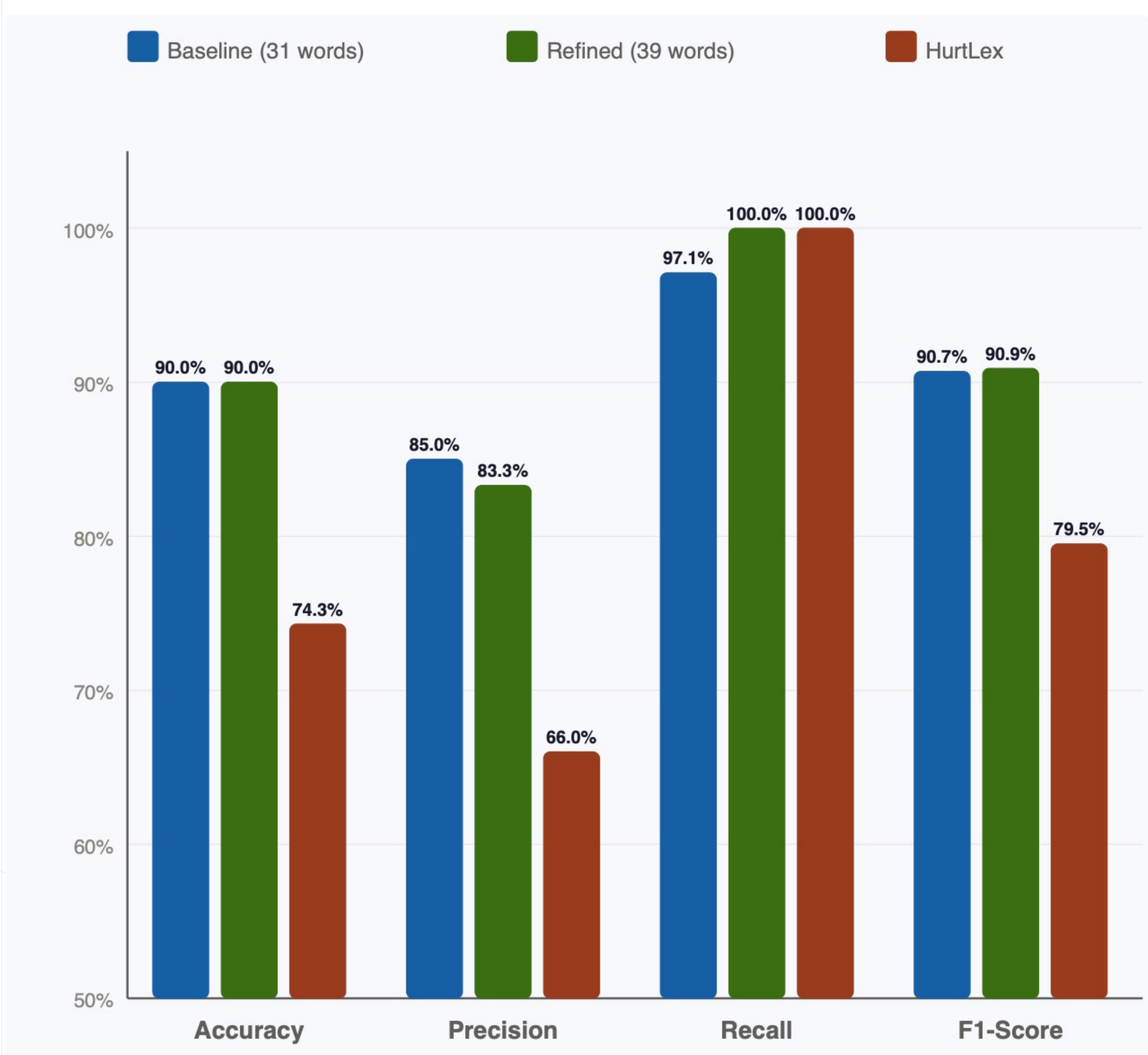


5 EVALUATION & RESULTS

System performance was evaluated using a labelled dataset of 70 sentences. The refined lexicon achieved 90% accuracy and 100% recall after optimisation. A comparison with the HurtLex lexical resource (Bassignana et al., 2018) produced lower accuracy due to increased false positives, confirming that a small domain specific lexicon provides reliable and interpretable detection.

90%	83.3%	100%	90.9%
ACCURACY	PRECISION	RECALL	F1-SCORE

Lexicon	Accuracy	Precision	Recall	F1-Score
Baseline (31 words)	90.0%	85.0%	97.1%	90.7%
Refined (39 words)	90.0%	83.3%	100%	90.9%
HurtLex	74.3%	66.0%	100%	79.5%



7 TECHNOLOGY STACK

Component	Technology
Language	Python 3.11
Framework	Flask
Database	SQLite
Frontend	HTML / CSS / JS
Authentication	Flask-Login
File Support	PDF, DOCX, CSV, TXT

6 CONCLUSION & FUTURE WORK

CONCLUSION

Talk Kindly demonstrates that a lightweight rule based NLP system can effectively detect offensive language while remaining transparent and easy to interpret. The refined lexicon achieved 90% accuracy and 100% recall, outperforming the general purpose HurtLex lexical resource.

- Expand the lexicon to cover more expressions and contextual variations
- Integrate machine learning methods to improve contextual understanding
- Extend support for multilingual offensive language detection
- Deploy as a real-time browser-based moderation tool

8 REFERENCES

- Bassignana, E., Basile, V. and Patti, V., 2018, December. Hurtlex: A multilingual lexicon of words to hurt. In Proceedings of the fifth Italian conference on computational linguistics (CLiC-it 2018) (pp. 52-57).
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